

July 5–9, 2026
Montreal, Canada

FSE 2026 Program



Sponsored by: ACM SIGSOFT

Supported by: Huawei, Bombardier, Intact Financial Corporation, Alipay, Uber, Google, JetBrains, Tourisme Montreal, Startupfest, the U.S. National Science Foundation, ACM-W Montreal Chapter, and SEMLA

General Chairs:
Shin Hwei Tan and Foutse Khomh

Conference Venue: Concordia SGW Campus @Downtown Montreal

The talks will be scheduled in rooms across two main buildings:

- [John Molson \(MB\) Building](#) (most talks) which is located at 1450 Guy St, Montreal, Quebec H3H 0A1.



- [Henry F. Hall Building](#) (keynote talks and plenary sessions) which is located at 1455 Blvd. De Maisonneuve Ouest, Montreal, QC H3G 1M8.



Location for Lunches:

- Lunches will be served in the Grey Nuns Residence (one of the recommended student accommodations), which is located at [1190 Guy Street](#).

Refer to [Campus Map](#) for more details:

Location

1450 Guy St, Montreal, Quebec H3H 0A1
Montreal
Canada

Program Overview					
Room	SUNDAY 5th July 2026	MONDAY 6th July 2026	TUESDAY 7th July 2026	WEDNESDAY 8th July 2026	THURSDAY 9th July 2026
	MORNING / SLOT 1 / BEFORE COFFEE BREAK				
	Sunday Monday 9:00- 10:30 - 90 minutes		Tuesday 8:30-10:30 - 120 min	Wednesday Thursday 8:30-10:00 - 90 minutes	
H 110			Welcome (30)+ Keynote 1 (40) + Keynote 2 (40)	SIGSOFT Impact Award Talk (25) + FSE research track distinguished paper award Talk (25) + Keynote 3 (40)	Test of Time Award Talk (25)+ FSE Distinguished artifact, Industry track distinguished paper & Reviewer Awards Ceremony (25)+ Keynote 4 (40)
MB 1.210		AIWare	AIWare		
MB 2.210		SSBSE (one day)			
MB 2.430		RSE (half day)			
MB 2.435		Tutorial 3 (180 min)	Tutorial 13 (180 min)		
MB 3.210		HumanAISE (one day)	New Faculty Symposium (one day)		
MB 3.270		DISE (one day)	Doctoral Symposium (one day)		
MB 3.430		PROMISE (one day)			
MB 3.435		SecDev (two days)	SecDev (two days)		
MB 3.445		DevOpsSustain (half day)	FaSE4Games (half day)		
MB 5.215					
MB 9A		IntersectionalitySE (half day)			
MB 9B					
MB 9C					
MB 9D			QSE-NE (half day)		
MB 1, MB 3, MB9	COFFEE BREAK (30 minutes)				
	Sunday, Monday 10:30-11:00		Tuesday 10:30-11:00	Wednesday, Thursday 10:00-10:30	
	MORNING / SLOT 2 / AFTER COFFEE BREAK, BEFORE LUNCH				
	Sunday, Monday 11:00-12:30 90 min		Tuesday 11:00-12:30 90 min	Wednesday, Thursday 10:30-12:30 - 120 min	
H 110				Industry Mentoring Symposium	
MB 1.210		AIWare	AIWare [90 min]	Testing 2 [120 min]	Testing 3 [120 min]
MB 2.210		SSBSE (one day)	LLM 4 SE 1 [90 min]	Code review 1 [120 min]	Developers [110 min]
MB 2.430	Tutorial 1 (90 min)	RSE (half day)	Library and Product Line [90 min]	Requirement and specification [120 min]	Root cause analysis [120 min]
MB 2.435	Tutorial 3 (180 min)	Tutorial 13 (180 min)	Mobile Apps 1 [90 min]	Cloud [120 min]	Services and Systems [120 min]
MB 3.210	HumanAISE (one day)	Doctoral Symposium (one day)	Testing 1 [90 min]	Repo-level code generation [110 min]	Code generation 2 [120 min]
MB 3.270	DISE (one day)	New Faculty Symposium (one day)	Porgram repair [90 min]	Code generation 1 [100 min]	Code and LLM 2 [100 min]
MB 3.430	PROMISE (one day)		Bug detection and localization [70 min]	Fuzzing 1 [100 min]	Fuzzing 3 [100 min]
MB 3.435	SecDev (two days)	SecDev (two days)	Fairness, green and sustainability [90 min]	Security 2 [100 min]	Software security [110 min]
MB 3.445	DevOpsSustain (half day)	FaSE4Games (half day)	Security 1 [90 min] with remote	Empirical 1 [120 min]	Empirical 2 [120 min]
MB 5.215			Program analysis 1 [70 min]	Comprehension and maintenance [120 min]	LLM 4 SE 6 [120 min]
MB 9A					
MB 9B	IntersectionalitySE (half day)	AIWare Competition 1	Dependency [90 min]	Student Res. Competition [120 min]	Release Engineering [120 min]
MB 9C					
MB 9D		QSE-NE (half day)	Benchmarking [90 min]	SEET 1 [120 min]	SEET 4 [120 min]
Grey Nun	Lunch at Grey Nun		Lunch at Grey Nun & Lunch Meetings		
MB 2.210			FSE Steering Committee (SC) meeting	TOSEM Editorial Board Meeting	FSE 2027 Organization Committee Meeting
MB 3.210			TSE Editorial Board Meeting	FSE 2027 PC	
H 110			Industry Mentoring Symposium		
MB 1.210		AIWare	AIWare [90 min]	LLM 4 SE 4 [90 min]	LLM 4 SE 7 [90 min]
MB 2.210	LLMSC (half day)	SSBSE (one day)	LLM 4 SE 2 [90 min]	LLM 4 SE 5 [90 min]	Code review 2 [90 min]
MB 2.430	Tutorial 2 (180 min)	Tutorial 4 (90 minutes)	Debugging [90 min]	Software Tests [90 min]	GUI Testing [90 min]
MB 2.435	Tutorial 5 (180 min)	Tutorial 10 (90 min)	Smart contract security [70 min]	Fuzzing 2 [90 min]	Failures [90 min]
MB 3.210	HumanAISE (one day)	Doctoral Symposium (one day)	SE and AI 1 [90 min]	SE and AI 2 [90 min]	SE and AI 3 [90 min]
MB 3.270	DISE (one day)	Doctoral Symposium (one day)	Agents [90 min]	Code and LLM 1 [90 min]	Code translation and transformation 2 [90 min]
MB 3.430	PROMISE (one day)		MSR 1 [90 min]	Program analysis 2 [90 min]	MSR 2 [90 min]
MB 3.435	SecDev (two days)	SecDev (two days)	Test generation 1 [90 min]	Performance and energy [90 min]	Verification 2 [90 min]
MB 3.445	LLMTrust (half day)	SEGA (half day, perfer full)	Logging [80 min]	Smart cotracts [60 min]	Vulnerability 2 [80 min]
MB 5.215			Requirement and design [90 min]	Anomaly and failure 1 [90 min]	Anomaly and failure 2 [90 min]
MB 9A					
MB 9B	SEE-AIT (half day)	AIWare Competition 2	SIGSOFT 50	Student Res. Competition [90 min]	Demystifying bugs [90 min]
MB 9C					
MB 9D	LLMSC (half day)	SE4ES-ASQAP (half day)		SEET 2 [90 min]	SEET 5 [90 min]
MB 1, MB3, MB9	COFFEE BREAK (15:30-16:00)				
	AFTERNOON / SLOT 4 / AFTER COFFEE BREAK				
	Sunday, Monday 16:00 - 18:00 120 min		Tuesday 16:00 - 17:30/17:40 90/100 min	Wednesday 16:00 - 17:00 60 min	Thursday 16:00 - 17:00 60 min
H 110					
MB 1.210		AIWare	AIWare [90 min]	FSE Debate	Closing+ SRC award+ FSE 2027
MB 2.210	LLMSC (half day)	SSBSE (one day)	LLM 4 SE 3 [90 min]		
MB 2.430	Tutorial 2 (180 min)	Tutorial 7 (90 min)	Verification 1 [80 min]	Broadening Participation Panel	
MB 2.435	Tutorial 5 (180 min)	Tutorial 9 (90 minutes)	Synthesis [80 min]		
MB 3.210	HumanAISE (one day)	Doctoral Symposium (one day)	Code LLM [90 min]	Mobile Apps 2 [60 min]	
MB 3.270	DISE (one day)	New Faculty Symposium (one day)	Vulnerability 1 [80 min]	UI [60 min]	
MB 3.430	PROMISE (one day)		Offensive security [80 min]	Process [60 min]	
MB 3.435	SecDev (two days)	SecDev (two days)	Code similarity and search [80 min]	Compiler [60 min]	
MB 3.445	LLMTrust (half day)	SEGA (half day, perfer full)	Robustness [70 min]	Quantum [50 min]	
MB 5.215			Test generation 2 [90 min]		
MB 9A					
MB 9B	SEE-AIT (half day)	AIWare Competition 3	SIGSOFT 50	Student Res. Competition [60 min]	
MB 9C					
MB 9D	LLMSC (half day)	SE4ES-ASQAP (half day)		SEET 3 [60 min]	
			Tuesday 17:30 - 18:30 60 min		
MB 9 ABCD			TOWN HALL		
MB 1.210					
			FSE Reception at St Jax	FSE Banquet at Grand Quai	

	Keynotes
	Paper presentation sessions
	Panel sessions
	Special event
	Workshop or colocated events

FSE26 Exclusive Events & Activities

Exclusive events and activities provided by our supporters includes (Advance Sign-up Required):

- Startupfest (July 8–10)
- Bombardier Tour (July 9 lunch time)
- Mila Tour (July 9 3.30pm)

FSE26 Plenary Events & Panels

Plenary events and panels include:

- *Broadening Participation Panel* by Mariam Barry and Ronnie de Souza Santos
- *Industry Mentoring Symposium* by Ahmed E. Hassan and Satish Chandra
- *FSE Debate: “The Future of Software Engineering in the Age of AI”* by Abhik Roychoudhury and Baishakhi Ray
- *SIGSOFT 50 Celebration* by Marsha Chechik

FSE 26 Reception & Banquet

FSE Reception on 7 July, 2026

Participants are responsible for arranging their own transportation to the St Jax (an Anglican church which also hosts many different events) which is 5 min walk from MB building.

Address: 1439 Rue Sainte-Catherine O, Montreal Quebec H3G 1S6 (Entry at the Rue Bishop)

Time and Date: 18:40-21:00 (7 July, 2026)

FSE Banquet on 8 July, 2026

Participants are responsible for arranging their own transportation to the Grand Quai (AML Cruise). There are several convenient ways to get there:

- Walking – Approximately a 40-minute walk from the MB Building.
- Public Transit – Accessible by metro and bus.
- Uber or Driving – Ride-share services and personal vehicles are also convenient options.

Address: 200 rue de la Commune Ouest (Grand Quai)

Banquet event schedule:

5.20 PM - Walking from MB building (if you would like to go with Student Volunteers)

6:00 PM – Boarding

7:00 PM – Cruise departure and meal service

10:30 PM – Return to dock and disembarkation

Keynotes

Tuesday 7 July FSE Opening Keynote 1: Benoit Baudry

Université de Montréal

Title: Punking Up Dependency Hell



Software does not exist in the void. A software application in any sector, of any size and complexity, relies on third-party dependencies. They provide reusable features, as well as test, build and deployment facilities. Born from the best software engineering practices of modularization and reuse, software dependencies also offer a fertile ground for abuses, from heartbleed to shai hulud, activism through protestware, or unnecessary code bloat. Development teams who used to handle a dependency hell, now need to manage the risks of a broader, richer network of dependencies, known as the software supply chain. In this talk, we discuss the latest research that keeps the software supply chain flame alive.

Bio: Benoit Baudry is a Professor in Software Engineering at the Université de Montréal and holds the Canada Research Chair in Reliable Software Supply Chain. Prior to joining Université de Montréal, he was a research scientist at INRIA (France) from 2004 to 2017, and a Professor at the KTH Royal Institute of Technology (Sweden) from 2017 to 2023. His work focuses on automated program analysis for observability, testing and randomization. He performs experimental research, for which he favors exploring code execution over code on disk.

Tuesday 7 July FSE Keynote 2: Dickson Tsai

Anthropic

Title: Harness Design for Increasingly Capable Models: How Newer Claude Code Features Shape Agent Trajectories



A coding agent's harness, the software that mediates the agent's interaction with the world, determines how much of a model's capability translates into delivered engineering work. As models have grown more capable, Claude Code's harness has evolved alongside them. This talk is a retrospective on Claude Code's public releases since the start of the year, detailing the design considerations behind each and the aggregate patterns they reveal. We frame each agent session as a trajectory: a search through the user's environment that accumulates knowledge and enacts changes within it. Through that lens, the year's launches sort into two moves: letting trajectories run further per unit of human attention (auto mode, routines, remote control, worktrees), and managing what each trajectory is conditioned on (memory,

multi-agent code review). Meanwhile, increased model capability strengthens the trajectories themselves. We conclude by examining how to orient these trajectories toward desired outcomes. Once step-by-step correctness is no longer the constraint, the binding problems become reconciliation—merging many independently correct trajectories into one repository—and validation—making it cheap for humans to review and revise intent for agents that are competent but not clairvoyant. Both can be software engineering problems, and we offer them to this audience as open questions.

Bio: Dickson Tsai is a Member of Technical Staff at Anthropic, where he has worked on Claude Code since shortly after its launch. He pioneered hooks and built the foundations for skills support—Claude Code's core extensibility primitives. Before Claude Code, he worked on pre-training data research at Anthropic and spent

six years as a growth engineer on Google Search. He is currently interested in the right extension points for developer teams to run richer coding agent trajectories.

Wednesday 8 July FSE Keynote 3: Sheila McIlraith

University of Toronto

Title: Towards Building Safer AI Systems



Advances in AI will transform many aspects of our lives and society, presenting immense opportunities but also posing significant risks and challenges. For many, large language models, chatbots, and agentic systems will serve as the main points of interaction with AI in the short-term, their benefits and risks increasingly well publicized. For computer scientists—practitioners, researchers, and educators—advances in AI are transforming the way we develop and deploy software, accelerating research, and prompting a rethinking of how we effectively educate students. The tension between measured, safe adoption of AI and expeditiously seizing its potential economic and societal benefits is palpable, with many western nations reporting low levels of trust among their citizens, and governments struggling

to identify appropriate governance and regulatory structures. In this talk I'll discuss some of the challenges to building and deploying “safe” AI systems, and potential pathways to safer adoption. To this end, I'll provide a brief technical deep dive into several areas where our decades of experience in formal methods and the development of safety-critical systems are inspiring methods to address the unique challenges of building safer AI systems—from synthesis, to verification, testing, auditing, and monitoring.

Bio: Sheila McIlraith is a Professor in the Department of Computer Science at the University of Toronto, a Canada CIFAR AI Chair (Vector Institute), and an Associate Director and Research Lead at the Schwartz Reisman Institute for Technology and Society. McIlraith is the author of over 150 scholarly publications in the areas of knowledge representation, automated reasoning, and machine learning. Her work focuses on AI sequential decision making, broadly construed, through the lens of human-compatible AI. McIlraith is a fellow of the Association for Computing Machinery (ACM), a fellow of the Association for the Advancement of Artificial Intelligence (AAAI), and a Schmidt Sciences AI2050 Senior Fellow. She is a member of the Canadian AI Safety Institute (CAISI) Research Council and Chair of the One Hundred Year Study of AI (AI100). McIlraith and coauthors have been recognized with a number of honours for their scholarly contributions including the 2011 SWSA Ten-Year Award, the ICAPS 2022 Influential Paper Award, and the 2023 IJCAI-JAIR Best Paper Prize.

Thursday 9 July FSE Keynote 4: Mary Shaw

Carnegie Mellon University

Title: Correctness, confidence, and context: Framing software assurance in the AI age



Software engineering has a complicated relationship with “correctness”. We recognize the challenges of full formal rigor as well as many required properties beyond functional correctness. Although we satisfice in practice, we are still stuck in the mindset that we could reason our way to correctness, if only we had enough information.

Generative AI has introduced a new dimension to assurances: its foundation is statistical rather than formal. Traditional software engineering establishes confidence through rigorous reasoning, domain knowledge and expert

judgment. In contrast, generative AI's results are sophisticated predictions, in Valiant's words “probably approximately correct”. This inherently limits assurances about the results are to probabilistic assertions.

Further, the nuances and implicit associations that guide human judgment are not accessible to its training sets, so that tacit knowledge cannot be incorporated in its models.

We have many approaches for developing assurances that a software system does what it's expected to do, though most of them focus on the specification of the code rather than the requirements for the system, let alone fitness for purpose. We have failed to develop a systematic understanding of the relative merits of the various approaches. I hope that generative AI will finally force us to tackle this.

To that end, I will challenge us to think systematically about our assurance techniques. We need ways to make informed, reasoned choices about cost-effective combinations of approaches to developing confidence in our systems.

We call ourselves software engineers. Let's act like engineers

Bio: Mary Shaw is the Alan J. Perlis University Professor of Computer Science in the Software and Societal Systems Department at Carnegie Mellon University. She has made fundamental contributions to an engineering discipline for software through developing data abstraction with verification (with W. Wulf and R. London), establishing software architecture as a major branch of software engineering (with D. Garlan), designing innovative curricula supported by two influential textbooks, and helping to found the Software Engineering Institute at Carnegie Mellon. She now works in software design. She has received the United States' National Medal of Technology and Innovation, the ACM SIGSOFT Outstanding Research Award (with David Garlan) and the IEEE Computer Society TCSE's Lifetime Achievement, Distinguished Educator, and Distinguished Women in Software Engineering Awards. She is an elected Fellow and Life Member of the ACM, the IEEE, and the American Association for the Advancement of Science.

Tutorials and Technical Briefing

ID	Title
1	Introduction to Causal Inference for Software Engineering
2	Generating Test Inputs and Interactions with the Fandango Fuzzer
3	Evaluating and Improving the Quality of LLM-Generated Code
4	Monitoring and Diagnosing Software Energy
5	Practical Software Performance Analysis on Linux Systems
7	Scaling Code Assistants Without Retraining with Model Merging: Capability Fusion & Sustainable Adaptation
9	Taming the Stochastic Machine: Observability and Runtime Governance for Agentic AI Systems
10	Technical Briefing on Beyond A/B Testing - Orthogonal and Experimental Designs
13	From Digital Twins to the Twinning Paradigm (What Every Software Engineer Should Know)

Workshops

13 co-located workshops includes:

- The 2nd International Workshop on Autonomous System Quality Assurance and Prediction with Digital Twins (ASQAP)
- The 1st International Workshop on Software Engineering for Generative Art (SEGA)
- Foundations of Applied Software Engineering for Games (FaSE4Games)
- International Workshop on Responsible Software Engineering (RSE)
- International Workshop on DevOps for Sustainability (DevOpsSustain)
- International Workshop on Large Language Model Supply Chain Analysis (LLMSC)
- International Workshop on Software Engineering for the GenAI Transformation (SEE-AIT)
- Intersectionality and Software Engineering (IntersectionalitySE)
- International Workshop on Software Engineering for and with Trustworthy LLMs (LLMTrust)
- International Workshop on Data Intensive Software Engineering (DISE)
- International Workshop on Software Engineering for Engineering Simulations, and Simulation Engineering for Engineering Software (SE4ES)
- Human-Centered AI for Software Engineering (HumanAISE)
- Third International Workshop on Quantum Software Engineering: The Next Evolution (QSE-NE)

FSE-AIWARE Joint Competition

3 FSE-AIWARE Joint Competitions includes:

- Agentic Based Python Dependency Resolution
- The Poisoned Chalice of LLM Evaluation
- Intelligent Coding Assistant Enhanced with Memory

Co-located Conferences

Co-located Conferences includes:

- ACM International Conference on AI-powered Software (AIWARE)
- ACM Secure Development Conference (SecDev)
- International Conference on Predictive Models and Data Analytics in Software Engineering (PROMISE)
- Symposium on Search-Based Software Engineering (SSBSE)

FSE 26 Sponsors and Supporters

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SEMLA
Trustworthy Engineering of AI Software





Ant Group

Ant Group traces its roots back to Alipay, which was established in 2004 to create trust between online sellers and buyers. Over the years, Ant Group has grown to become one of the world's leading open Internet platforms.

Through technological innovation, we support partners in providing inclusive, convenient digital financial services to consumers and SMEs. In addition, we have been introducing new technologies and products to support the digital transformation of industries and facilitate collaboration. Working together with global partners, we enable merchants and consumers to make and receive payments and transact around the world.

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FROM RESEARCH IDEAS TO UBER-SCALE IMPACT



JETBRAINS

JetBrains Research advances the state of the art in computer science and software engineering through practical research and innovative solutions.

Our Talks at FSE 2026

Human-AI Experience in Integrated Development Environments: A Systematic Literature Review

July 7th • 14:30 • MB 3.210



Does In-IDE Calibration of Large Language Models work at Scale?

July 8th • 14:20 • MB 3.270



CoRenameAgent: A Human-In-The-Loop, Multi-Agent Tool for Coordinated Rename Refactoring

July 9th • 15:00 • MB 3.270



Configurable AI Coding Assistants: Designing For Developers Who Like to Be in Control

HumanAISE workshop • July 5, 2026



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